

About Reality Representation Standard - (RRS)

Governing How Reality Becomes Represented Inside a Simulation

Represented Inside a Simulation

A simulation never contains reality itself.

It contains a representation of selected reality.

That representation may simplify physical systems, abstract human behavior, approximate environments, compress complex interactions, or reproduce only those parts of reality considered relevant to the simulation purpose.

Reality Representation Standard (RRS) governs this transformation.

Its purpose is to ensure that organizations can distinguish between:

what exists in reality

and:

what the simulation actually represents.

Canonical Definition

Reality Representation Standard (RRS) defines the governance conditions under which the aspects of reality relevant to a governed Simulation Object, Purpose, Scope, Applicability Context, and required Simulation Depth & Precision are selected, abstracted, represented, resolved, simplified, bounded, and documented within a simulation so that the simulated representation remains explicitly distinguishable from the reality it claims to represent.

Why RRS Exists

Every simulation must make choices.

It must decide: which real-world characteristics matter, which behaviors must be represented, which interactions require detail, which elements may be simplified, which phenomena require approximation, and which real-world conditions cannot be represented sufficiently.

Without explicit governance, these choices can disappear inside the model.

The simulation may then appear more realistic, complete, or broadly applicable than the representation actually supports.

RRS prevents that.

Governed Space

Simulation Reality Representation

RRS governs the methodological bridge between:

relevant reality

and:

its simulated representation.

It does not determine what reality actually is.

That remains within the Operational Reality governance space.

RRS determines how the selected reality relevant to the simulation inquiry is translated into representable form.

Governance Flow

Governed Simulation Inquiry → Reality Selection → Reality Abstraction → Representation Fidelity → Representation Resolution → Simplification & Approximation → Representation Limitations → Governed Reality Representation

This representation basis then supports:

Simulation Assumptions → Model & Environment → Inputs & Data → Scenario Design → Simulation Execution

What RRS Establishes

RRS helps an organization determine:

Reality Selection — which real-world characteristics actually need to appear in the simulation.

Abstraction — which complexity may be reduced without destroying material behavior.

Fidelity — how faithfully relevant reality must be represented.

Resolution — how finely the simulation must distinguish relevant states, events, spaces, times, and interactions.

Simplification — which real-world complexity may legitimately be reduced.

Approximation — where direct representation is impossible or unnecessary.

Representation Limitations — where the simulated reality differs materially from the reality it is intended to represent.

Reality Is Not the Model

One of the most important distinctions in SIMULOS® is:

Reality ≠ Simulation Representation

and:

Simulation Representation ≠ Simulation Model

Reality is the external or intended world.

Reality Representation defines which characteristics of that world must enter simulation.

The model is the technical mechanism later used to implement that representation.

This distinction allows organizations to change models or simulation technologies without losing the underlying representation requirements.

Fidelity Must Follow Purpose

Higher fidelity is not automatically better.

A simulation used for early design exploration may require relatively simple representations.

A simulation supporting high-consequence autonomous deployment may require significantly stronger representation of: system behavior, interactions, environmental variation, failure conditions, timing, dependencies, and operational states.

RRS therefore does not ask:

How realistic can we make the simulation?

It asks:

How faithfully must the relevant reality be represented for this particular simulation purpose?

Simplification Is Legitimate

Simulation necessarily simplifies reality.

RRS does not attempt to eliminate simplification.

It makes simplification explicit and governable.

An organization may simplify: geometry, environmental variation, human behavior, component detail, system interactions, or physical processes

where those simplifications do not remove characteristics material to the simulation inquiry.

The key principle is:

Simplification must not silently remove the reality the simulation is intended to examine.

Representation Limitations

No representation is complete.

RRS therefore requires material representation limitations to remain visible.

Examples may include: incomplete human behavioral representation, simplified environmental dynamics, proxy data, incomplete physical models, limited failure behavior, uncertain system interactions, simplified network conditions, or static representation of a dynamic system.

These limitations become important when organizations interpret simulation outputs later.

Business Value

For organizations investing heavily in simulation, RRS provides a structured way to answer a difficult but fundamental question:

Does the simulated world contain enough of the real world to support the question we are asking?

This helps organizations identify: unnecessary simulation complexity, missing material reality, hidden simplifications, insufficient fidelity, inappropriate abstraction, simulation gaps, and situations where high numerical precision creates false confidence.

It also improves communication between:

engineering teams, simulation teams, decision-makers, safety teams, auditors, validators, and operational users

because all parties can refer to the same governed Reality Representation Basis.

Relationship to Operational Reality

The boundary with ORA™ is explicit.

ORA™ asks:

What is actually happening in Operational Reality?

RRS asks:

How must the relevant parts of that reality be represented inside this simulation?

RRS therefore does not duplicate Operational Reality governance.

It creates the simulation-specific representation layer between actual reality

and the simulation architecture.

Relationship to Validation

RRS also does not determine that a simulation is valid.

A simulation may possess a carefully governed Reality Representation and still require formal validation.

VALIDOS® governs whether the relevant simulation, simulation output, evidence, or claim can legitimately satisfy a validation purpose.

RRS provides one important part of the simulation basis that such validation may later examine.

Practical Governance Test

Before model construction proceeds, an organization should be able to answer:

Which aspects of reality must this simulation represent?

Which aspects may be abstracted?

Which elements require high fidelity?

Which elements may use lower fidelity?

What resolution is actually necessary?

What has been simplified or approximated?

Which important parts of reality cannot currently be represented sufficiently?

Could any of those limitations materially change the meaning of the simulation outputs?

If those questions cannot be answered, simulation complexity alone cannot establish representation quality.

Architectural Position

RRS is the third Parent Standard of SIMULOS® — Simulation Governance Architecture.

The architectural progression is:

Simulation Object

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Simulation Purpose & Scope



Reality Representation

SOS establishes:

What exactly are we simulating?

SPSS establishes:

Why are we simulating it and what must the inquiry cover?

RRS establishes:

Which aspects of reality must now be represented and how must that representation remain bounded by abstraction, fidelity, resolution, simplification, approximation, and known limitations?

The next governance question becomes:

Which assumptions are necessary to make this representation and the resulting simulation possible?

Foundational Principle

A simulation can legitimately represent reality only to the extent that the reality elements material to its purpose are explicitly selected, appropriately abstracted, sufficiently represented, and bounded by their known representational limitations.

Canonical Closing Statement

Reality Representation Standard (RRS) governs the transformation from relevant reality into simulated representation by defining what must be represented, what may be abstracted or simplified, how much fidelity and resolution are required, and which limitations must remain visible throughout the simulation lifecycle. By separating reality from its simulated representation, RRS prevents technical detail, numerical precision, or model complexity from being mistaken for real-world equivalence and establishes the governed reality basis upon which assumptions, models, environments, inputs, scenarios, execution, outputs, and later simulation-to-reality correlation can be constructed.

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